

Swapping AttnGrounder's Attention for a Brain-Inspired Associative Memory (BDH)

Progress Update: Method, Results, and Open Questions

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The Task: Visual Grounding

Visual grounding (Talk2Car): given an image and a natural-language command, localize the single object the command refers to.

“Turn left behind the silver car” → one bounding box.

Scenes with **multiple objects of the same class** are the harder case — the model has to use the language to tell them apart, not just detect objects.

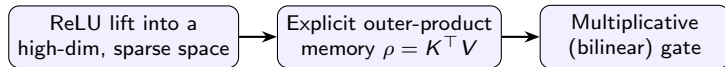
Baseline model

AttnGrounder (Mittal et al., 2020) — single-stage, Darknet-53 + BiLSTM/GloVe, a lightweight *visual-text attention* module fuses the two modalities before a YOLO head.

Why Look at BDH?

Dragon Hatchling (BDH) — Kosowski et al., Pathway, 2025

A recently published sequence architecture whose working memory is an **explicit, growing outer-product associative memory** ($\rho = K^\top V$), updated with Hebbian-style read/write, rather than compressed into a fixed-size state the way SSMs such as Mamba do.



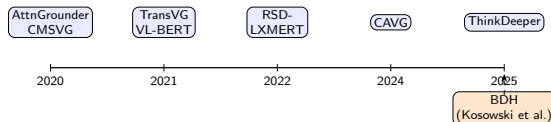
The question this project asks

BDH has, as far as we can tell, not previously been evaluated on any cross-modal (vision + language) task. We wanted to see whether it can be substituted for an established, years-refined attention mechanism in a grounding model, and what happens to accuracy, parameter count, and compute cost when we do.

A more specific hypothesis about why BDH's memory might help is discussed later, once the main comparison has been presented.

Prior Approaches to Talk2Car Grounding

Model	Year	Backbone / fusion	Test AP50
AttnGrounder	2020	CNN (Darknet-53) + BiLSTM	61.32
CMSVG (best / lightweight)	2020	EfficientNet + RoBERTa / DistilBERT	68.61 / 67.8
TransVG	2021	CNN + BERT, transformer self-attn fusion	65.83
VL-BERT (Talk2Car)	2021+	CNN + BERT, joint pretraining	70.03
RSD-LXMERT	2022	LXMERT	72.64
CAVG	2024	not detailed in our reading	74.62
ThinkDeeper	2025	ViT+BERT (BLIP-init) + world model/hypergraph	76.64



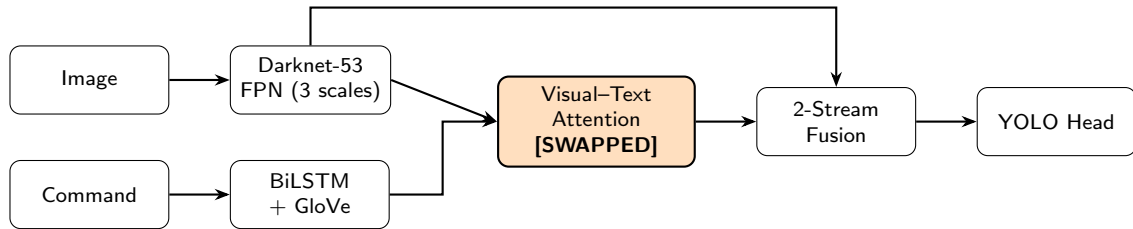
Reading the table

Accuracy tends to track backbone/pretraining tier more than fusion-mechanism design — VL-BERT, RSD-LXMERT, CAVG and ThinkDeeper all add a pretrained transformer backbone that AttnGrounder and CMSVG don't have. BDH (2025) and the current Talk2Car leaderboard entry (ThinkDeeper, also 2025) exist at the same time but, as far as we found, had never been connected before this project — only AttnGrounder/CMSVG predate the test-server closure (see “Why Val, Not Test” later).

BDH and Related Memory Architectures

- BDH's original paper reports experiments on **language modeling and machine translation only**, at GPT-2-like parameter scales (10M–1B). We did not find vision or cross-modal experiments described in the paper text we read (its Appendix E code listing did not extract cleanly for us, so we cannot fully rule out something there, but nothing in the main text describes one).
- **State-space models (e.g. Mamba)** compress context into a fixed-size hidden state; BDH's memory instead grows with the input. This contrast motivates testing BDH on a task that plausibly rewards keeping distinguishing information around rather than compressing it — a hypothesis we treat as untested, not an established property of BDH.
- We read the papers collected for this project (BDH; AttnGrounder; ThinkDeeper; IRRA — a cross-modal relation-reasoning paper for a different task, text-to-image person retrieval) and ran an ad hoc search (via Consensus) around adjacent ideas such as distractor-based hard negatives. We found no prior work applying BDH to visual grounding or any cross-modal task.
- Not a systematic literature review — absence of a result in our search is evidence of a gap, not proof no such work exists.

Architecture: Swapping One Module

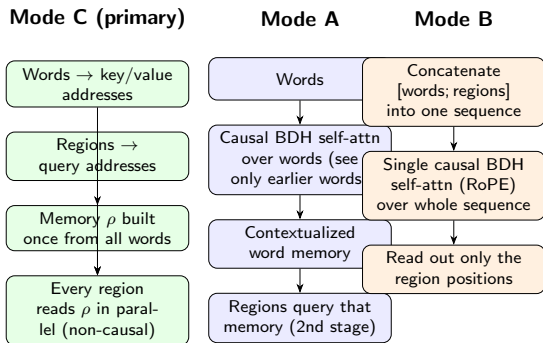


Baseline attention: softmax cross-attention (region queries $\times$ word keys), scalar gate β .

BDH module: symmetric or causal-context kernel (3 modes, see next slide); ReLU-sparse addressing; explicit outer-product memory; multiplicative bilinear gate. Same output shapes, so it can be substituted without changing the rest of the network.

How Modes A, B, C Fit Into the Architecture

BDH was designed for causal, single-sequence language modeling. Grounding has no inherent order between “words” and “regions,” so we tried three different ways of adapting the mechanism; only Mode C has been trained and evaluated end-to-end so far (Modes A and B are implemented and ran, see Results).



Mode C keeps BDH’s memory mechanism but drops the causal mask and RoPE (no ordering assumption between words and regions). Modes A and B keep more of BDH’s original causal design, closer to how it was used for language modeling.

Keeping the Comparison Fair

- **Single component swap:** backbone, text encoder, YOLO head, losses, hyperparameters — all identical between variants.
- **Fusion trim:** BDH fuses only $[f_{visu}, f_{lang_attn}]$ (2 streams) instead of the original 3, to offset most of the module's added parameters.
- **Same training recipe for all variants:** A100-40GB, bf16, 100 epochs, Adam (lr 10^{-4} , backbone lr/10), batch 14, poly LR schedule.

Variant	Params	vs. baseline
Baseline (AttnGrounder, unmodified)	75.84M	—
BDH (any mode, single-head)	76.63M	+0.79M (+1.0%)

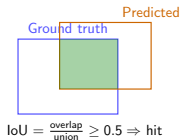
The backbone dominates parameter count either way — this is a fusion-mechanism comparison, not a capacity comparison.

The Metrics, Explained

AP50 (our main number)

A prediction counts as **correct** if the predicted box overlaps the ground-truth box by

Intersection-over-Union ≥ 0.5 . AP50 = % of commands localized correctly.



Params

Total trainable parameters — a proxy for model size / compute cost. Lower is “lighter.”

Ambiguous vs. Unambiguous (Step 6)

The stratification used to test our hypothesis —
ambiguous: ≥ 1 other same-class object in frame;
unambiguous: referred object is the only one of its class. Built from nuScenes 3D metadata — see next slide.

Step 6: Stratifying the Val Set

Talk2Car's own annotations only record the *referred* object — not what else is in the scene. To test the hypothesis we need to know, per image, how many **other** objects share the referred object's class.

Data pipeline built for this

Join Talk2Car commands $\rightarrow$ nuScenes v1.0-trainval metadata (front-camera 3D annotations) via the sample token $\rightarrow$ count same-class objects visible in CAM_FRONT.

Bucket	Definition	Count (val, $n=1163$)
Ambiguous	≥ 1 other object, same class	830
Unambiguous	referred object is unique in class	333

Ablation Study — Full Results (Multi-Seed)

Baseline and Mode A were reseeded after a single-seed run did not replicate; the table below reports multi-seed means, not that original single run:

Variant	AP50 (all) <i>n</i> =1163	AP50 (ambig.) <i>n</i> =830	AP50 (unambig.) <i>n</i> =333	vs. base (all)
Baseline (mean, 2 seeds)	64.92	64.04	67.12	—
Mode A (mean, 3 seeds)	65.09	63.62	68.77	+0.17
Mode B	60.53	58.19	66.37	−4.39
Mode C	64.06	62.17	68.77	−0.86
Mode E (C + growing mem.)	64.75	62.77	69.67	−0.17

Variance matters when reading this table

Mode A's 3 seeds spanned **64.14–66.09** (all); baseline's 2 seeds were tighter (**64.89–64.92**). BDH's spread looks larger than baseline's given how few seeds we have — a single encouraging seed (the original +0.61 ambiguous result) did *not* replicate on reseed, so we treat any single-seed number cautiously.

Reading the Ablation Results

Observations

Mode A's mean (3 seeds) is slightly above baseline's mean (2 seeds) on aggregate AP50 (+0.17); Mode E improves on its own base kernel (+0.69 over Mode C) and is above baseline on unambiguous scenes. Taken together, a mechanism with no prior cross-modal testing performs *comparably to* an attention mechanism that has had years of task-specific refinement, on this one benchmark and this one architecture.

- These differences are small relative to the seed-to-seed variance shown on the previous slide, so we read them as directional rather than conclusive with only 2–3 seeds per variant.
- Not every kernel formulation works — Mode B (60.53) is a clear regression — so this is not “any BDH variant helps,” it appears to depend on the specific design choice.

Why does this hold up, mechanistically, if it does? We had a specific hypothesis going in — see “Examining the Motivating Hypothesis” later for what the data actually shows.

Efficiency: Lighter in Params, Heavier in Compute

	Params	Layers	GFLOPs	Inference	FPS
Baseline	75.84M	183	43.49	2.13ms	~470
Mode A	76.63M	184	50.04	2.39ms	~418
Delta	+1.0%	+1	+ 15.1%	+ 12.5%	-11.1%

Three numbers, not one

Params alone looks flattering (+1%). GFLOPs tells a different story: lifting into the high-dimensional sparse space ($n = \text{mult} \times d = 1024$) is compute-heavy at *each* of the 3 FPN scales, even though the lift matrices themselves are shared and cheap in parameter count. That shows up as a measured +12.5% increase in inference time. We think all three numbers belong in the writeup together.

Where We Stand — Same-Class Comparison

ThinkDeeper (Liao et al., Nov 2025, ViT+BERT+world-model+hypergraph, **76.64** AP50) is #1 on the Talk2Car leaderboard, but its own Table 1 shows a bracket structure by backbone class:

Model	Backbone	AP50 (test)	Params
AttnGrounder (paper)	CNN	61.32	75.84M
Ours: BDH Mode A	CNN	65.09 (val, mean 3 seeds)	76.63M
CMSVG	CNN (EfficientNet)	68.61	—
VL-BERT	CNN+BERT	70.03	~118M
RSD-LXMERT	LXMERT	72.64	~210-230M
ThinkDeeper	ViT+BERT+...	76.64	not stated

A more realistic target than “reach 76.64”

Reaching ThinkDeeper’s number would need a full transformer/VLM backbone swap — a different project. A same-bracket target that stays within a fusion-mechanism study: **CMSVG’s 68.61**, the top result in our same-class (CNN, non-VLM) bracket. We are not there yet.

We are also smaller than the transformer-class models (60–70% of VL-BERT’s params, roughly a third of RSD-LXMERT’s) and much smaller than VLM-based ones (7B–72B params).

Where This Differs From Prior Work

Based on the reading and searching described under “Related Work” — framed as “we did not find this elsewhere,” not as an unfalsifiable “first ever.”

BDH applied to cross-modal grounding

BDH’s own paper evaluates language modeling and translation only. We did not find it applied to vision, to cross-modal fusion, or to Talk2Car anywhere in the literature we reviewed — as far as we can tell, this is a first attempt at a cross-modal grounding application.

Three non-causal kernel adaptations

BDH was designed causal/autoregressive (language has sequence order; grounding does not). Modes A/B/C are our own adaptations of BDH’s associative-memory mechanism to a non-causal, bimodal setting, not a direct reuse of the original code.

- **Quantitative ambiguity stratification:** nuScenes-joined same-class-object counting on Talk2Car’s val split (not shipped with the dataset) — tests the hypothesis at the sub-population level, not just in aggregate.
- **The fusion trim as a confound control:** holds params near-parity (+1%) with baseline, so the mechanism swap is close to the only variable changing — several papers we read change backbone, mechanism, and data together, which makes it harder to attribute gains to any one factor.

A Note on Methodology: Why Val, Not Test

- Talk2Car’s test-set ground truth has **never been publicly released** — the only historical evaluation path was an Alcrowd-hosted server (ECCV 2020 C4AV challenge).
- That server **closed to submissions in August 2020**. We found no evidence of a live replacement.
- AttnGrounder and CMSVG were published while that server was still open, so their test numbers are plausibly server-verified. Papers reporting Talk2Car “test” scores since then (TransVG 2021, RSD-LXMERT 2022, CAVG/ThinkDeeper 2024–2025) all point to that same closed evaluation path — their numbers look most likely **self-reported**, and we could not verify how they were produced.

Our choice

We report **val-set AP50 only**, explicitly caveated against other papers’ self-reported test numbers. This is less flattering than quoting a test number, but it is the number we can stand behind — directionally informative, not a strict apples-to-apples comparison with every row in the tables above.

The Hypothesis Behind This Project

“Grounding = in-context retrieval”

Our starting idea: architectures with an explicit associative memory might have an edge specifically when a query has to be matched against **multiple similar candidates** in context — framing that as a retrieval problem rather than a compression problem (BDH's memory is explicit and growing; SSMs like Mamba compress to a fixed-size state instead).

Prediction: swapping in BDH should raise AP50 *disproportionately* on scenes with several same-class candidate objects (“ambiguous,” $n=830$) relative to scenes with only one (“unambiguous,” $n=333$) — tested via the nuScenes-joined stratification described earlier (Step 6). *This was the project's original motivating question; it came before the results, not after, and we think it is worth being explicit about that.*

Result: Hypothesis Not Supported

Across every configuration tested

Mode A ($\times 3$ seeds), B, C, E — all 6 BDH data points — none clearly exceed baseline's own observed ambiguous-scene range (64.04 mean, 63.37–64.70 across seeds). The disproportionate ambiguous-scene advantage the hypothesis predicted does not show up consistently across the ablation.

- Where BDH *does* come out ahead (Mode A's aggregate, Mode E's unambiguous gain), it is concentrated in the *opposite* bucket from what the hypothesis predicted — the easier, single-candidate scenes, not the harder disambiguation cases.
- Mode B — the variant closest to BDH's original causal, sequential design — is the weakest performer, not the strongest, which argues against a simple “more associative memory means more retrieval benefit” story.

Why report a negative result at all?

It answers the question that motivated the project, even though the answer is “no.” It also tells us the viability result above isn't coming from the mechanism we originally expected, which narrows down what to look at next.

Additional Experiments — A Retrospective

Four follow-up experiments after the ablation study, each carried through to a real conclusion rather than left open:

- **Tversky + Focal loss** (replacing BCE on the β /attention-map supervision): **negative** — 64.03, below Mode A's 65.09. The aux loss only carries 10% weight in the total objective, so there may not be enough leverage for it to matter.
- **Cross-scale growing memory (Mode E)**: a real *within-mechanism* improvement (+0.69 over its own base kernel, Mode C) — but still no advantage over baseline on ambiguous scenes specifically. See Results.
- **Offline/frozen pretrained text encoder** (DistilBERT, replacing GloVe+BiLSTM): **substantial regression** — 59.98 vs. baseline's 64.92. A LayerNorm fix (targeting a real BERT-scale mismatch) recovered only +0.52. Our best guess at the root cause: a fully *frozen* encoder has almost no capacity to adapt to this task. We **paused** this track rather than abandoning it — partial fine-tuning or domain-adaptive pretraining seem like more promising next steps than another architectural tweak.
- **Efficiency metrics**: done — see earlier slide.

Scoped out for now: YOLOv7/YOLO26 backbone swap, BDH stacking, a BDH→attention hybrid, a distractor-contrastive loss — plausible ideas we did not chase further once the AttnGrounder phase had a complete answer we felt comfortable reporting.

Cross-Architecture Check: TransVG

The question

Everything so far is one architecture (AttnGrounder, 2020, narrow region $\leftrightarrow$ word cross-attention). Does BDH's viability hold on a structurally different design, or is it specific to that integration point?

TransVG (Deng et al., ICCV 2021) — same recipe, one variable changed

BDH swapped into TransVG's V-L fusion **self**-attention (6 stacked layers over ~ 421 tokens: 1 reg + 20 text + ~ 400 visual) — a structurally different role than AttnGrounder's narrow cross-attention. We verified a healthy baseline (55.20% val accu) and correct plumbing (an overfit-8-sample probe cleanly memorizes).

Result: BDH plateaus at roughly half the baseline

Baseline 55.20% vs. **BDH 26.23%** val accu (best epoch, same epoch for both) — the same overfit probe (softmax MHA reaches 1.0 accu on 8 fixed samples; BDH plateaus at ~ 0.5 – 0.625) suggests this is a real, consistent gap, not a bug.

A plausible explanation, not a confirmed one: TransVG forces one pooled associative memory from all ~ 421 tokens as both keys and values; visual tokens outnumber text/reg tokens 20:1 and may dominate it, with no per-token normalization to protect the signal, compounded across 6 stacked layers — untested, see Future Work.

Possible Next Steps

None of the following is committed work, just what currently looks worth trying based on where the results leave off:

- **Test the TransVG explanation directly:** add per-token normalization to BDH's self-attention memory read and see whether that narrows the gap to baseline, rather than concluding self-attention roles are unsuitable for BDH based on one architecture.
- **Revisit the frozen text-encoder track:** partial DistilBERT fine-tuning or domain-adaptive continued pretraining instead of a fully frozen encoder — paused, not abandoned, and the more promising of the two remaining fixes identified.
- **Distractor-contrastive auxiliary loss** (queued, not yet implemented): use the nuScenes-derived same-class distractor locations already built for Step 6 as hard negatives in a margin loss on β — untested, may or may not help.
- **Multi-head BDH:** everything reported here uses a single head; multi-head is an unexplored capacity/parameter trade-off.
- **Independent test-set verification:** given the val/test caveats discussed earlier, contacting the Talk2Car maintainers about research-use test-set access would let comparisons happen on a currently-verifiable split.
- **Backbone modernization** (e.g. a newer detector than Darknet-53): scoped out so far because it changes more than one variable at once; if pursued, it would need to apply identically to baseline and BDH variants to stay a fair comparison.

Overview of Findings

Three separate findings, reported here together rather than leading with any one of them:

On AttnGrounder's narrow cross-attention: a scoped positive result

A memory architecture published under a year ago, with no prior cross-modal validation that we found, achieves AP50 comparable to — and, on the aggregate mean across our seeds, slightly above — an established, years-refined softmax attention mechanism, with no task-specific tuning of the BDH mechanism itself (+1% params, +15% compute). A real but *narrow* result, specific to this integration point.

The mechanism behind it is not what we predicted

Across every AttnGrounder configuration tested (3 kernel modes, multi-seed on the best one, the growing-memory extension), BDH shows **no disproportionate advantage on ambiguous, multi-candidate scenes** — the “grounding = in-context retrieval” idea that motivated this project. Reported as a genuine negative result, not a footnote.

It does not generalize naively to every attention role

Ported to TransVG's *self*-attention (dense, one pooled memory over ~ 421 visual-dominated tokens) instead of AttnGrounder's narrow cross-attention, BDH plateaus at **roughly half** the baseline (26.23% vs. 55.20% val accu), which an overfit probe suggests is a real limitation, not a bug. Viability observed on AttnGrounder appears *role-specific*.

What differs from prior work: a first cross-modal application of BDH, as far as we found; three non-causal kernel adaptations of an inherently causal mechanism; a reusable ambiguity-stratification protocol; and a partial characterization of which attention role suits BDH's memory. Negative findings belong alongside the positive one, not underneath it.

Summary

- **Main result:** on AttnGrounder's narrow cross-attention, a memory architecture published under a year ago, with no prior cross-modal testing that we found, performs comparably to established softmax attention (mean AP50 65.09 vs. 64.92 across seeds), with no task-specific tuning of the mechanism.
- Built a validated, drop-in BDH swap (3 kernel modes + a cross-scale extension) for AttnGrounder, a from-scratch baseline reproduction, and a nuScenes-based ambiguity-stratification pipeline.
- Efficiency, reported together rather than cherry-picked: +1% params, +15% GFLOPs, +12.5% inference time.
- **Multi-seed testing:** the original motivating hypothesis — a specific edge on ambiguous, multi-candidate scenes — did not hold up under seed-tested scrutiny. An encouraging single-seed result did not replicate.
- **Cross-architecture check (TransVG):** the viability finding does *not* generalize naively — BDH self-attention over TransVG's dense, visual-dominated tokens plateaus at roughly half the baseline (26.23% vs. 55.20%). We believe this is a real boundary, not a bug, though we have not yet tested our leading explanation for it (see Future Work).
- Several concrete next steps identified (Future Work) — none of them started yet.

Thank you — questions and corrections welcome.